

SET 6: PRODUCT SALES ANALYSIS

Dataset: Monthly Product Sales

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

PROGRAM 1: Grouped Bar Chart of Product Sales

Experiment: Data Visualization using R

Question: Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Aim

To create a grouped bar chart in R that shows the January, February and March sales of each product for the first quarter, with all chart elements labelled.

Dataset

The Monthly Product Sales dataset given above is used.

Procedure

1. Create the Monthly Product Sales data frame in R.
2. Select the January, February and March columns and convert them into a matrix, using the product names as row names.
3. Use `barplot()` with `beside = TRUE` and the transposed matrix `t(sales_matrix)` to draw separate bars for January, February and March for each product.
4. Add the title, X-axis label, Y-axis label and the Month legend inside `barplot()`.
5. Use `text()` to display the sales values as data labels on top of the bars.

```
# Dataset
sales <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
```

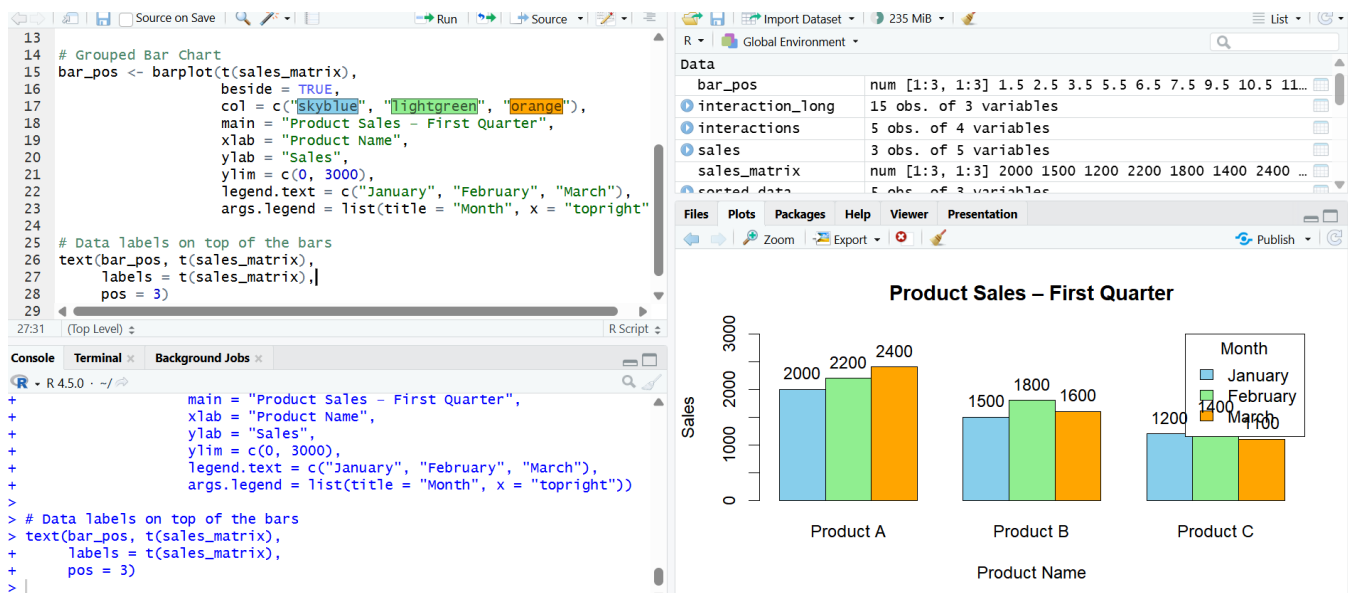
```
# Matrix of monthly sales (rows = products, columns = months)
sales_matrix <- as.matrix(sales[, c("January", "February", "March")])
rownames(sales_matrix) <- sales$Product_Name

# Grouped Bar Chart
bar_pos <- barplot(t(sales_matrix),
  beside = TRUE,
  col = c("skyblue", "lightgreen", "orange"),
  main = "Product Sales - First Quarter",
  xlab = "Product Name",
  ylab = "Sales",
  ylim = c(0, 3000),
  legend.text = c("January", "February", "March"),
  args.legend = list(title = "Month", x = "topright"))

# Data labels on top of the bars
text(bar_pos, t(sales_matrix),
  labels = t(sales_matrix),
  pos = 3)
```

Expected Output

A grouped bar chart titled Product Sales – First Quarter is displayed. Product Name is on the X-axis and Sales is on the Y-axis. Each product has three bars (January, February and March) in different colours, the Month legend is displayed, and the sales values are shown on top of the bars.



Output 1: Product Sales – First Quarter (Grouped Bar Chart)

Result

The grouped bar chart was successfully created in R to visualize the first-quarter sales of each product, with the chart title, axes, Month legend and data labels labelled.

PROGRAM 2: Stacked Area Chart of Overall Sales Trend

Experiment: Data Visualization using R

Question: Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Aim

To create a stacked area chart in R that represents the overall sales trend of all products over January, February and March.

Procedure

1. Load the ggplot2 package (install it once using `install.packages("ggplot2")`) and create the Monthly Product Sales data frame.
2. Arrange the data in long format with the columns Product, Month and Sales, and set the Month order as January, February, March using `factor()`.
3. Use `ggplot()` with Month on the X-axis, Sales on the Y-axis and Product as the fill colour, and add `geom_area()` to draw the stacked areas, so that each product contributes to the total monthly sales.
4. Add the title, axis labels and the Product Name legend using `labs()`.

```
library(ggplot2) # install.packages("ggplot2") if not installed

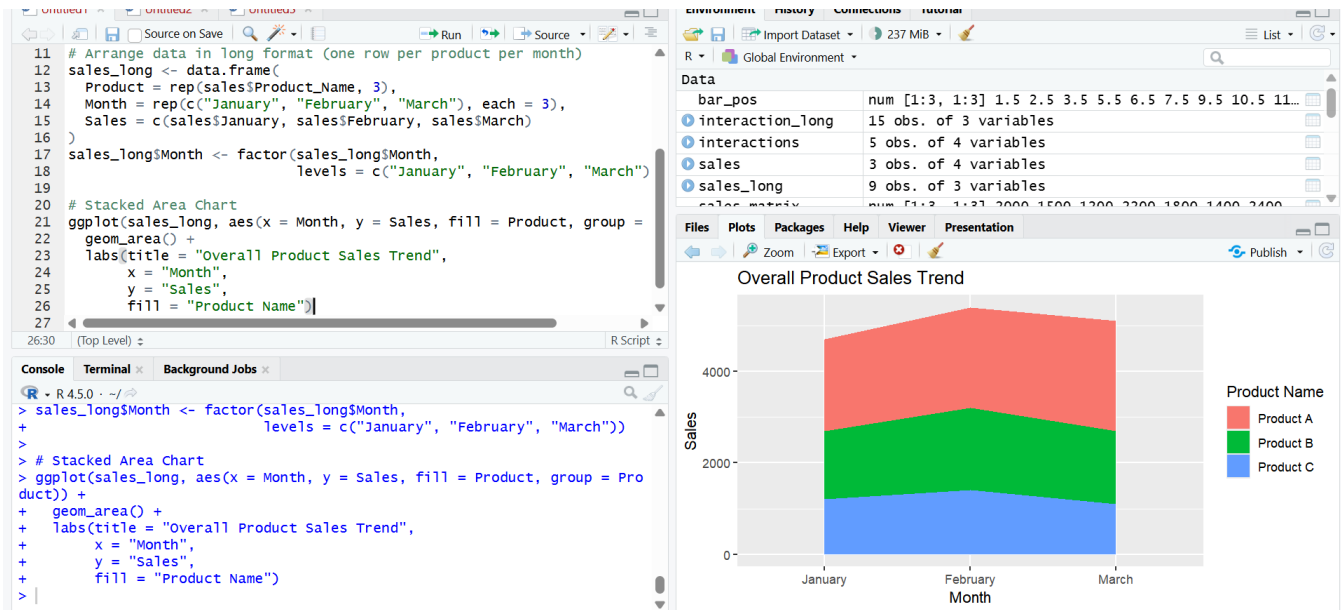
# Dataset
sales <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

# Arrange data in long format (one row per product per month)
sales_long <- data.frame(
  Product = rep(sales$Product_Name, 3),
  Month = rep(c("January", "February", "March"), each = 3),
  Sales = c(sales$January, sales$February, sales$March)
)
sales_long$Month <- factor(sales_long$Month,
                          levels = c("January", "February", "March"))

# Stacked Area Chart
ggplot(sales_long, aes(x = Month, y = Sales, fill = Product, group = Product)) +
  geom_area() +
  labs(title = "Overall Product Sales Trend",
       x = "Month",
       y = "Sales",
       fill = "Product Name")
```

Expected Output

A stacked area chart titled Overall Product Sales Trend is displayed with Month on the X-axis and Sales on the Y-axis. Product A, Product B and Product C are shown as three stacked coloured areas with a Product Name legend. It shows each product's contribution to the overall sales across the three months.



Output 2: Overall Product Sales Trend (Stacked Area Chart)

Result

The stacked area chart was successfully created in R. It shows the contribution of each product to the overall sales across January, February and March.

PROGRAM 3: Monthly Product Sales Table

Experiment: Data Visualization using R

Question: Build a table to show the monthly sales data for each product. Label the table elements.

Aim

To build a table in R that shows the monthly sales of each product, with the table elements labelled.

Procedure

1. Create the data frame with the columns Product_Name, January, February and March.
2. Use cat() to display the table title Monthly Product Sales Table.
3. Use print() with row.names = FALSE to display the table.

```
# Dataset
sales_table <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

# Table title
cat("Monthly Product Sales Table\n\n")

# Display the table
print(sales_table, row.names = FALSE)
```

Table elements: the title is Monthly Product Sales Table, the row header is Product_Name (Product A, Product B, Product C), the column headers are January, February and March, and the sales values are the numbers inside the table.

Expected Output

The table titled Monthly Product Sales Table is displayed in the R console as shown below:

Product Name	January	February	March
Product A	2000	2200	2400
Product B	1500	1800	1600
Product C	1200	1400	1100

```
1 # Dataset
2 sales_table <- data.frame(
3   Product_Name = c("Product A", "Product B", "Product C"),
4   January = c(2000, 1500, 1200),
5   February = c(2200, 1800, 1400),
6   March = c(2400, 1600, 1100)
7 )
8
9 # Table title
10 cat("Monthly Product Sales Table\n\n")
11
12 # Display the table
13 print(sales_table, row.names = FALSE)
14 |
```

14:1 (Top Level) R Script

Console Terminal Background Jobs

```
R R 4.5.0 ~/  
# Table title  
cat("Monthly Product Sales Table\n\n")  
Monthly Product Sales Table  
  
# Display the table  
print(sales_table, row.names = FALSE)  
Product_Name January February March  
Product A 2000 2200 2400  
Product B 1500 1800 1600  
Product C 1200 1400 1100
```

Output

3: Monthly Product Sales Table

Result

The monthly product sales table was successfully built in R, showing the sales of each product for January, February and March with the table title, row header, column headers and sales values labelled.

PROGRAM 4: Interactive Tableau Dashboard

Experiment: Data Visualization using Tableau

Question: Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Aim

To develop an interactive Tableau dashboard that combines the grouped bar chart, the stacked area chart and the monthly sales table for exploring the product sales data.

Procedure

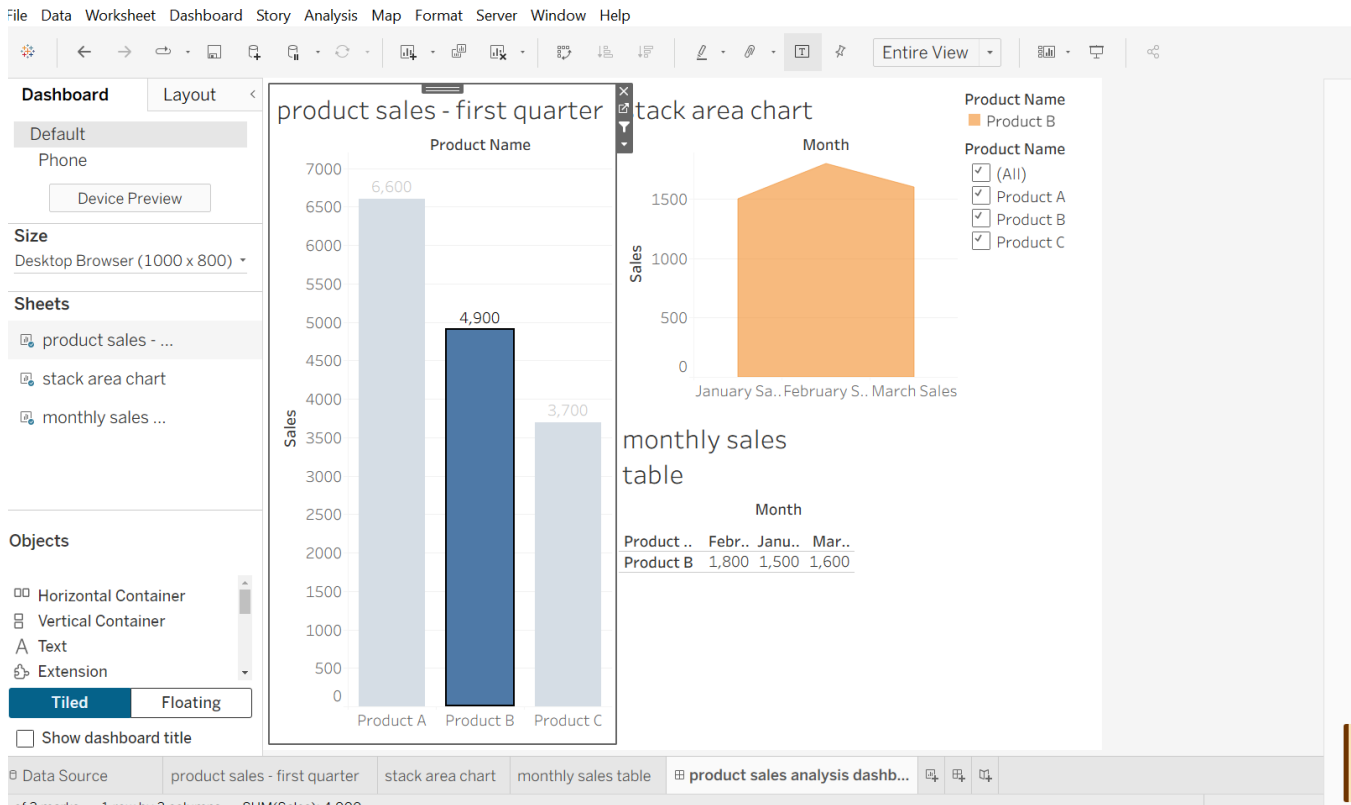
1. Save the Monthly Product Sales dataset as an Excel (.xlsx) or CSV file. Open Tableau, click Microsoft Excel (or Text file) under Connect, select the file and click Open. Drag the sheet to the canvas.
2. Reshape the data: click January Sales, hold Ctrl and click February Sales and March Sales, then right-click and choose Pivot. Rename Pivot Field Names to Month and Pivot Field Values to Sales.
3. Create the grouped bar chart: click Sheet 1 and rename it Grouped Bar Chart. Drag Product Name to Columns, Sales to Rows, and Month to Columns beside Product Name. Select Bar in the Marks card, drag Month to Color, and turn on Show Mark Labels (Label > Show mark labels). Set the title to Product Sales – First Quarter.
4. Create the stacked area chart: add a new worksheet and rename it Stacked Area Chart. Drag Month to Columns and Sales to Rows, change the Marks type to Area, drag Product Name to Color, and make sure Analysis > Stack Marks is On. Set the title to Overall Product Sales Trend.
5. Create the table: add a new worksheet and rename it Monthly Sales Table. Drag Product Name to Rows, Month to Columns and Sales to Text in the Marks card. Set the title to Monthly Product Sales Table.
6. For all three worksheets, right-click Month, choose Sort > Manual, and keep the order January, February, March.
7. Click the New Dashboard icon at the bottom of the screen. A blank dashboard opens with the list of worksheets on the left side.
8. Right-click the dashboard tab, choose Rename, and type Product Sales Analysis Dashboard.
9. Set the dashboard size: in the left pane under Size, select Fixed size (for example, 1200 x 800) or Automatic so that all three components fit on one screen.
10. Drag the Grouped Bar Chart worksheet and drop it at the top-left part of the dashboard.
11. Drag the Stacked Area Chart worksheet and drop it on the right side of the grouped bar chart (top-right part).
12. Drag the Monthly Sales Table worksheet and drop it at the bottom of the dashboard, below both charts.
13. Arrange the components neatly by resizing their edges, so that all three are fully visible without overlapping.
14. Add the dashboard title: click Dashboard on the top menu and tick Show Title. Double-click the title and type Product Sales Analysis Dashboard.
15. Add filters: go to each worksheet and drag Product Name and Month to the Filters shelf (select all values and click OK). On the dashboard, click a chart, click the drop-down arrow at its top-right corner, choose Filters, and tick Product Name and Month so that the filter boxes appear.
16. Apply the filters to all charts: click the drop-down arrow on each filter box, choose Apply to Worksheets, and select All Using This Data Source. Selecting a product (for example, Product A) or a month (for example, February) now updates the bar chart, the area chart and the table together.
17. Add interaction between the charts: click Dashboard on the top menu and choose Actions. Click Add Action, select Filter, set the Source Sheet as Grouped Bar Chart and the Target Sheets as Stacked Area Chart and Monthly Sales Table, set Run action on to Select, and click OK. (Alternatively, use the Use as Filter funnel icon on a chart.)
18. Test the dashboard by clicking a bar or selecting a filter value. The other charts and the table change to show the selected product or month. Click a blank area (or choose All in the filter) to return to the full view.
19. Make the dashboard clean and readable by removing unnecessary legends and empty spaces, and keep the layout given below.

Dashboard Layout

- Top: Dashboard title – Product Sales Analysis Dashboard
- Left/top: Grouped Bar Chart
- Right/top: Stacked Area Chart
- Bottom: Monthly Sales Table
- Filters: Product Name and Month

Expected Output

A dashboard titled Product Sales Analysis Dashboard is displayed with the grouped bar chart on the top-left, the stacked area chart on the top-right and the monthly sales table at the bottom, along with the Product Name and Month filters. Selecting a product or month updates all three components for interactive exploration of the sales data.



Output 4: Product Sales Analysis Dashboard

Result

The Product Sales Analysis Dashboard was successfully created in Tableau by combining the grouped bar chart, the stacked area chart and the monthly sales table, with Product Name and Month filters for interactive exploration of the sales data.